

Homework Assignment: Classifier Evaluation

1 Learning Objectives

Upon completing this assignment, you should be able to:

- Calculate and interpret common metrics for evaluating binary classifiers.
- Explain the effects of class imbalance and classification thresholds on classifier evaluation.
- Construct and interpret ROC and Precision–Recall curves.
- Apply cross-validation to select hyperparameters while maintaining an independent test dataset.
- Build a leakage-resistant machine-learning pipeline for regularized logistic regression.
- Evaluate and interpret the generalization performance of a logistic regression classifier using biological data.

2 Problems

2.1 Problem 1: Conceptual Understanding of Classifier Evaluation

A research group develops a logistic regression model to predict whether a patient has a disease (1 = disease, 0 = no disease) using five blood biomarkers. The dataset contains 1,000 patients, of whom 100 have the disease and 900 do not. The logistic regression model outputs an estimated probability

$$\hat{P}(Y = 1 \mid X).$$

Using a classification threshold of 0.50, evaluation on a held-out dataset gives the following confusion matrix:

	Predicted Negative	Predicted Positive
Actual Negative	820	80
Actual Positive	30	70

2.1.1 Performance metrics

Identify TP, TN, FP, and FN. Calculate each of the following metrics:

- Accuracy
- Sensitivity/Recall
- Specificity
- Precision
- F1 score

Show your calculations.

2.1.2 Accuracy and class imbalance

Is accuracy alone an adequate measure for evaluating this classifier? Explain your answer in the context of the class distribution and disease prediction.

2.1.3 Effect of changing the classification threshold

Suppose the classification threshold is decreased from 0.50 to 0.30. Without calculating exact values, explain what will generally happen to:

- TP
- FN
- FP
- TN
- Recall
- Precision

Why does changing the classification threshold create a tradeoff between precision and recall?

2.1.4 ROC versus Precision–Recall curves

- Explain the difference between an ROC curve and a Precision–Recall (PR) curve.
- Which curve would you pay particular attention to for this disease-prediction problem, and why?
- Your explanation should consider the class distribution rather than simply stating that one curve is “better” than the other.

2.1.5 Hyperparameter tuning and the test dataset

- The researchers want to tune the logistic regression hyperparameter C . They propose the following procedure:

Try

$$C = 0.001, 0.01, 0.1, 1, 10, 100$$

and report the test-set performance of whichever value of C produces the highest test-set AUC.

Explain what is wrong with this procedure.

- Describe a correct workflow. Be explicit about when the test dataset may be used.

2.1.6 Feature scaling and information leakage

- Explain why predictor scaling should be performed *inside* the cross-validation process rather than standardizing the entire dataset before cross-validation.
- Also explain why scaling is particularly important when regularization is used in logistic regression.

2.2 Problem 2: Evaluating a Logistic Regression Classifier Using Biological Data

In this problem, you will use the **Pima Indians Diabetes dataset**. The dataset contains clinical measurements from adult female patients of Pima Indian heritage and is commonly used to study the prediction of diabetes. Your goal is to develop and evaluate a logistic regression classifier that predicts whether a patient has diabetes. For this assignment, define the outcome as

$$Y = \begin{cases} 1, & \text{diabetes,} \\ 0, & \text{no diabetes.} \end{cases}$$

The dataset contains the following predictor variables:

- **Pregnancies:** Number of pregnancies
- **Glucose:** Plasma glucose concentration
- **BloodPressure:** Diastolic blood pressure
- **SkinThickness:** Triceps skin-fold thickness
- **Insulin:** 2-hour serum insulin
- **BMI:** Body mass index
- **DiabetesPedigreeFunction:** Diabetes pedigree function
- **Age:** Age in years

The outcome variable is **Outcome**, where 1 indicates diabetes and 0 indicates no diabetes.

2.2.1 Load and explore the dataset

Load the provided Pima Indians Diabetes dataset into a pandas DataFrame. Create:

- **X:** predictor variables
- **y:** binary outcome (**Outcome**)

Report the following:

- Number of samples
- Number of predictors
- Number and percentage of patients with diabetes
- Number and percentage of patients without diabetes

Briefly discuss whether the classes are balanced.

Important: In this dataset, a value of zero is not biologically plausible for some clinical measurements and should therefore be treated as missing. Replace zero values with missing values (**NaN**) for the following variables:

- Glucose
- BloodPressure
- SkinThickness
- Insulin
- BMI

Do **not** replace zero values for `Pregnancies`, because zero is a valid value for this variable. For each of the five variables listed above, report the number of values that are treated as missing.

2.2.2 Create development and test datasets

Split the complete dataset into:

- 80% development data
- 20% test data

Use a stratified split and set `random_state=42`. For example,

```
train_test_split(
    X,
    y,
    test_size=0.20,
    stratify=y,
    random_state=42
)
```

Call the resulting datasets:

```
X_dev, X_test, y_dev, y_test
```

Report the number of samples from each class in the development and test datasets. Explain why the test dataset should remain untouched during subsequent model development.

2.2.3 Construct a machine-learning pipeline

- Construct a `scikit-learn` Pipeline containing:

```
SimpleImputer → StandardScaler → LogisticRegression
```

Use the **median** of each predictor to impute missing values:

```
SimpleImputer(strategy="median")
```

Use L2 regularization for logistic regression.

- Explain why both `SimpleImputer` and `StandardScaler` should be placed inside the pipeline rather than imputing missing values and standardizing predictors using the complete dataset before cross-validation. In particular, explain how the pipeline helps prevent information leakage during cross-validation.

2.2.4 Select the regularization parameter using cross-validation

Consider the following candidate values of C :

`C_values = [0.001, 0.01, 0.1, 1, 10, 100]`

- Use **5-fold stratified cross-validation** on the **development dataset only** to select C .
- Use **AUC** as the primary model-selection metric.
- For each value of C , report the mean and standard deviation of the validation AUC values across the five folds.
- Complete a table similar to the following:

C	Mean of CV AUC	SD of CV AUC
0.001		
0.01		
0.1		
1		
10		
100		

- Plot

Mean of CV AUC versus C

using a logarithmic scale for the x -axis.

- Identify the selected value of C and explain how you selected it.

2.2.5 Train the final model

Using the selected value of C , fit the final pipeline using **all development data**. Do not use the test data during this step.

2.2.6 Evaluate the final classifier

Now evaluate the final model on the test dataset. Using a classification threshold of 0.50, report:

- Confusion matrix
- Accuracy
- Sensitivity/Recall
- Specificity

- Precision
- F1 score

Interpret **sensitivity** and **precision** in the biological context of diabetes prediction. For example:

A sensitivity of _____ means that _____% of patients with diabetes in the test dataset were correctly identified as having diabetes.

2.2.7 ROC analysis

Using the predicted probabilities for the test samples:

- Plot the ROC curve.
- Add the diagonal random-classifier reference line.
- Calculate AUC.
- Label the x -axis and y -axis appropriately.
- Explain in 2–3 sentences what the AUC tells you about the classifier.

2.2.8 Precision–Recall analysis

Using the same predicted probabilities:

- Plot the Precision–Recall curve.
- Calculate Average Precision (AP).
- Add the positive-class prevalence as the no-skill baseline.
- Explain what happens to precision and recall as the classification threshold changes.

2.2.9 Biological interpretation and model evaluation

Write a short paragraph addressing all of the following questions:

- Does the classifier appear to generalize well to previously unseen samples? Support your conclusion using the results of your analysis.
- If this classifier were intended as a diabetes screening tool, would false positives or false negatives be of greater concern? How would this consideration affect your choice of classification threshold?
- Why should you **not** go back and change C after seeing the test-set AUC, ROC curve, or Precision–Recall curve?

3 Submission Requirements

Submit the following:

- Your answers to Problem 1.
- One Jupyter notebook (`.ipynb`) for Problem 2. The Jupyter notebook should contain:
 - Clearly organized Python code
 - Appropriate comments
 - All requested numerical results
 - The cross-validation results table
 - The CV AUC versus C plot
 - The confusion matrix
 - The ROC curve
 - The Precision–Recall curve
 - Written interpretations in Markdown cells

All code cells should be executed and their outputs should be visible before submission.